LAB 2

SWITCH-CONTROLLED LED & CYCLIC BLINKING

Embedded Systems Laboratory • TM4C123GH6PM

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1. Objective

To configure the TM4C123GH6PM GPIO for switch and RGB LED interfacing and implement two programs. Program 1 uses the two switches to select the LED colour according to all four possible switch combinations. Program 2 provides user-controlled blinking, where SW1 selects the blinking rate and SW2 selects the LED colour cyclically.

2. Hardware & GPIO Configuration

GPIO Port F is enabled and configured for the onboard RGB LED and two push buttons. PF1, PF2 and PF3 are configured as digital outputs for the Red, Blue and Green LED channels respectively. PF4 (SW1) and PF0 (SW2) are configured as digital inputs with internal pull-up resistors. PF0 is a protected pin and is therefore unlocked and committed before configuration. Since the switches are active-low, a released switch reads logic 1 and a pressed switch reads logic 0.

Pin	Connected device	Role
PF0	SW2	Input + pull-up
PF4	SW1	Input + pull-up
PF1	Red LED	Output
PF2	Blue LED	Output
PF3	Green LED	Output

3. Program 1 — Switch-Controlled RGB LED

The program continuously polls the Port F input register and isolates PF4 and PF0 using bit masks 0x10 and 0x01. The detected switch combination is then mapped to an RGB output. The LED output codes are Red = 0x02, Green = 0x08, Blue = 0x04 and White = 0x0E. White is obtained by enabling all three RGB channels simultaneously.

SW1	SW2	Input state	LED
Pressed	Not pressed	0,1	Blue
Not pressed	Pressed	1,0	Green
Pressed	Pressed	0,0	White
Not pressed	Not pressed	1,1	Red

One Drive Link: - Part01

4. Program 2 — Switch-Controlled Blinking

The second program introduces two state variables, rate and colour. The rate index selects one of three software delay values: 1000 (slow), 500 (medium) and 250 (fast). The colour index selects Red, Green, Blue or White. SW1 increments the rate index and SW2 increments the colour index; when the last value is reached, the index wraps back to zero.

Control	Cyclic values	Function
SW1	1000 → 500 → 250 → 1000	Blinking rate
SW2	R → G → B → W → R	LED colour

To avoid multiple changes from one long button press, the program stores switch1Value and switch2Value as the previous input states. A selection changes only when the current input is 0 and the previous value was 1, corresponding to a released-to-pressed transition. The LED is driven ON for the selected delay, then turned OFF for the same delay. The switches are checked during both intervals so that a user input can be detected throughout the blink cycle.

One Drive Link: - Part02

5. Results, Observation & Conclusion

Program 1 produces the specified Red, Green, Blue and White outputs for all four switch combinations. Program 2 successfully changes the blinking speed using SW1 and the LED colour using SW2, with both parameters cycling through their fixed sets. The experiment demonstrates GPIO clock and pin configuration, digital input/output operation, active-low switch logic, bit-mask based input detection, RGB colour generation, software timing, state variables and edge-based switch detection. The required switch-controlled LED functionality was successfully implemented on the TM4C123GH6PM.